

CTF-GS: one compact foundation-feature Gaussian map

Frozen foundation heads supervise training; deployed readouts query the same compact scene memory as 2D features, 3D primitives, and 3D points.

Offline supervision

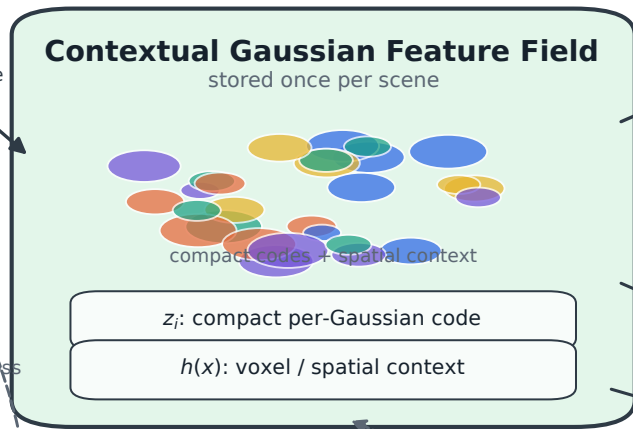
posed RGB views
+ 3DGS geometry

dense RADIO teacher
1280-D features

frozen task heads
SigLIP2 / DINO / SAM3

multiview primitive
registration

Stored compact map



Global readout heads

**Foundation-Space
Reconstruction**

View-Conditioned
Calibration

Support-Calibrated
Primitive Readout

Multi-Head
Foundation Consistency

Protocol evidence

LERF rendered-view OVS
2D heatmap / mask

LERF direct 3D OVS
primitive support

ScanNet point query
open-vocabulary points

Frozen-head probes
SigLIP2 / DINO / SAM3

Training-only constraints

- foundation-space reconstruction + frozen-head consistency
- visibility-aware support distillation from multiview registration
- geometry regularization + SAM3 mask-logit/boundary supervision

— inference
- - - training only

Dashed paths supervise training only; deployed readouts share the stored compact map.